

Muzaffar Khaydarov

Rising Junior, B.S. Computer Science & Mathematics — Class of 2028
662-550-0100 | mkhaydar@go.olemiss.edu | linkedin.com/in/muzaffar- | github.com/mountstorm

EDUCATION

University of Mississippi
Bachelor of Science in Computer Science, Minor in Mathematics
• GPA: 3.87/4.0 — Chancellor's Roll (all semesters) — Sally McDonnell Barksdale Honors College (Top 5%)
• Relevant Coursework: Data Structures & Algorithms, Computer Organization & Assembly Language, Object-Oriented Programming, Database Systems, Data Science, Discrete Mathematics, Linear Algebra, Statistics

Oxford, MS
Aug. 2024 – May 2028

EXPERIENCE

Software Engineering Intern
C Spire
• Own 2-3 repositories handling billing audit logic and application flows, shipping code changes (not just reviews) across production Perl and Python systems
• Handled 100+ weekly zero-payment errors in a production Perl billing flow by routing invalid records to a rejection file, preventing batch-wide failures and removing on-call manual intervention
• Built an environment-aware Python automation that queries flagged accounts from a relational database and auto-issues a ServiceNow (SNOW) ticket only when actionable issues exist, replacing manual review of 10–20 daily account emails and saving on-call operators 30 min/day

June 2026 – Present
Ridgeland, MS

Undergraduate Research Assistant — Distributed AI Systems
University of Mississippi (Prof. Feng Wang) & University of Memphis
• Implement split-inference systems in Python (PyTorch) that partition deep neural network execution across device, edge, and cloud to reduce latency and enable AI on memory- and GPU-constrained hardware
• Built a two-machine distributed inference pipeline (Mac head + Linux edge node) with a length-prefixed TCP protocol to transfer intermediate tensors, verifying output identical to single-machine inference (max error under 1e-4)
• Profiled all 22 AlexNet split points for compute vs. transmission cost across WiFi, LTE, and backhaul links to find the latency-optimal partition; validated a column-parallel (Megatron-style) layer split as mathematically equivalent

Jan. 2026 – Present
Oxford, MS

Computer Science Teaching Assistant
University of Mississippi
• Guide 49+ students through core CS fundamentals including object-oriented programming (Java), data structures, and algorithms through hands-on labs, debugging sessions, and code reviews
• Review and optimize 300+ programs weekly, improving code quality and implementation accuracy by 20%

Jul. 2025 – Present
Oxford, MS

Undergraduate Research Assistant
University of Mississippi — Thad Cochran Research Center
• Programmed industrial robots (Wilkata, Epson) using Python and C++ for autonomous pharmaceutical preparation, reducing contamination risk by 85% through automated drug delivery workflows
• Built data pipelines using pandas, NumPy, Matplotlib for collection, preprocessing, and visualization

Oct. 2024 – Apr. 2025
Oxford, MS

PROJECTS

NeuraBash
Python, Bash, OpenAI GPT-4, Ollama, Prompt Engineering
• Built a terminal-based AI coding assistant integrating GPT-4 and local LLMs (Ollama) for online and offline inference, applying prompt engineering to drive repository analysis and code explanation
• Engineered a modular CLI with a memory-efficient architecture, reducing RAM footprint by 30%

Sept. 2025 – Oct. 2025

ValueStop
EmoryHacks 2025 — React, Vite, Gemini 2.0 API, Google Maps API, Firestore
• Developed AI-powered shopping navigator optimizing multi-store routes using Gemini API and A* pathfinding
• Engineered two-tier data architecture with pre-scraped database (150 products, 5 stores) plus real-time web scraping using Playwright; collaborated with 3-person team using Git

Nov. 2025

TECHNICAL SKILLS

Languages: Java, Python, C++, C, JavaScript, TypeScript, Perl, SQL (PostgreSQL, MySQL), Bash
Frameworks & Tools: Spring Boot, Node.js, Next.js (React), Flask, LangChain, Vite, Git, Docker, Kubernetes, AWS (EC2), Firebase, ServiceNow, PyTorch, Ollama, Claude Code, Cursor
Systems & Platforms: Linux, Red Hat Enterprise Linux (RHEL), Bash scripting, Billing Systems, Production Support
Concepts: Microservices, Distributed Systems, REST APIs, Workflow Automation, Prompt Engineering, pandas, NumPy, pytest, Relational Databases, Algorithms, Edge Computing, Workload Partitioning
Certifications: AWS Generative AI Applications (Oct 2025), IBM Data Analysis Professional (Oct 2025)